

Md Yekra Rahman

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SUMMARY

- **Available for Summer 2026 Internship in Analog IC Design.**
- Experienced in transistor-level design of op-amps, LDOs, bandgap references, charge-pumps, differential amplifiers and gate driver - level shifter circuits using Cadence Virtuoso using gm/ID methodology.
- Skilled in block validation, presentations and documentation.

EDUCATION

PhD Student in Electrical Engineering	Fall 2025 – Present
University of Missouri - Columbia, MO, USA	
PhD Student in Electrical Engineering, Concentration: Circuits	Fall 2024 – Summer 2025
TxACE, University of Texas at Dallas, Richardson, TX, USA	GPA: 3.83/ 4.00
B.Sc. in Electrical and Electronic Engineering, Major : Electronics	April 2018 – May 2023
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh	GPA: 3.65/ 4.00

SKILLS

IC Design Tools: Cadence Virtuoso (schematic, simulation, layout supervision), CMOS layout familiarity (DRC/LVS, parasitic extraction), LTSpice

Lab & Validation Equipment/Software: Oscilloscope, Network Analyzer, Signal Generator, SKILL, MATLAB, Python

RESEARCH EXPERIENCE

Projects	Fall 2024 – Present
Design of an Operational Amplifier using IBM 130 nm CMOS technology	
• Designed a two-stage, Miller-compensated CMOS op-amp in IBM 130 nm using a small-signal and gm/ID-driven flow; sized in transistor-level and the layout aware compensation network to meet unity-gain bandwidth and phase-margin targets under a specified load and single-supply constraint. • Validated performance with dedicated testbenches for open-loop gain/GBW/PM, slew rate, output swing, CMRR, and power, ensuring all transistors remained in saturation and complying with the single ideal current-source requirement.	
DC–DC Buck Converter Design using TSMC 180 nm HV BCD process	
• Designed a 12 V to 3.3 V, 1 A, 2 MHz synchronous (half-bridge) buck in Cadence Virtuoso using the TSMC 180 nm HV BCD PDK; optimized MOSFETs via width sweeps for loss/efficiency tradeoffs. • Implemented gate driver with up/down level shifters, bootstrap, and dead-time; tuned edges and validated in ADE with 5.15% output ripple and 198 mA inductor ripple, verifying V_{GS}, level-shifter I/O, and bootstrap behavior.	
Hybrid Converter PCB: 4-Layer Design in Altium	
• Built a 4-layer hybrid-converter PCB from scratch in Altium: created custom symbol/footprint libraries, and generated Gerber and drill files. • Applied EMI-aware stackup and layout, top/bottom partitioned for power and control; inner GND and VDD planes; minimized switching-loop and return paths with close decoupling and domain separation for signal and power integrity.	

TEACHING EXPERIENCE

Graduate Teaching Assistant	Fall 2024 – Present
<i>University of Missouri - Columbia; University of Texas at Dallas</i>	
• Assisted in teaching CMOS IC Design and related courses, guiding students in Cadence Virtuoso and SPICE simulations for analog circuits. • Mentored undergraduate students in lab sessions, covering op-amp design, layout checks (DRC/LVS), and circuit validation using oscilloscopes and function generators.	
Lecturer	August 2023 – August 2024
<i>Eastern University</i>	<i>Dhaka, Bangladesh</i>

PUBLICATIONS

M. Y. Rahman and S. M. Mominuzzaman, "Exploring Lead Free Mixed Halide Double Perovskites Solar Cell," 2024 13th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2024, pp. 165-170, doi: 10.1109/ICECE64886.2024.11024609.